

SUSTAINABILITY REPORT

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Of course. Here is the STRUCTMIND AI Sustainability Report for the specified structural steel package.

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STRUCTMIND AI: Sustainability Report

Project Component: Structural Steel Package Total Tonnage: 220 tons (199.6 metric tons) Manufacturing Process: Comparative Analysis: Electric Arc Furnace (EAF) vs. Basic Oxygen Furnace (BOF) Coating System: Standard SSPC 3-Coat System (Assumed: Zinc/Epoxy/Urethane) Transportation: Truck Freight, 900 miles (1,450 km)

1. Embodied Carbon Estimate (A1-A3 Manufacturing)

This estimates the "cradle-to-gate" carbon emissions from raw material extraction through the manufacturing process. The choice of mill is the single most significant factor.

- Electric Arc Furnace (EAF) Mills:
 - kgCO₂e per ton: ~600 kgCO₂e / ton (0.6 tCO₂e / ton)
 - Total Project Embodied Carbon: 132,000 kgCO₂e (132 metric tons CO₂e)
- Basic Oxygen Furnace (BOF) Mills:
 - kgCO₂e per ton: ~2,200 kgCO₂e / ton (2.2 tCO₂e / ton)
 - Total Project Embodied Carbon: 484,000 kgCO₂e (484 metric tons CO₂e)

Analysis: Specifying steel from an EAF mill reduces the manufacturing-stage embodied carbon of this package by approximately 73% (a saving of 352 metric tons of CO₂e) compared to steel from a typical BOF mill. This is due to EAF's use of recycled scrap as its primary feedstock.

2. Recycled Content % Target

Recycled content is the primary driver of the embodied carbon difference.

- EAF Mills: The feedstock is almost entirely recycled scrap steel.

- Target Recycled Content: >90% (Typically 93-97%)
- BOF Mills: The feedstock is primarily virgin iron ore, with a smaller amount of scrap used for process control.
- Typical Recycled Content: 25-35%

Recommendation: To meet sustainability goals, mandate steel from EAF mills to maximize recycled content.

3. Transportation Carbon Estimate (A4 Transport)

This estimates the "gate-to-site" emissions from transporting the fabricated steel.

- Methodology:
- Total Ton-Miles: 220 tons * 900 miles = 198,000 ton-miles
- Emission Factor: ~0.16 kgCO₂e per ton-mile (Typical for heavy-duty truck freight)
- Calculation:
- 198,000 ton-miles * 0.16 kgCO₂e/ton-mile = 31,680 kgCO₂e
- Total Transportation Carbon: ~32 metric tons CO₂e

Analysis: While significant, transportation emissions are secondary to manufacturing emissions. Sourcing from a fabricator/mill closer to the project site is the most effective way to reduce this impact.

4. Coating VOC / Low-Emission Options

A standard 3-coat system (e.g., inorganic zinc primer, epoxy mid-coat, polyurethane topcoat) provides excellent corrosion protection but can have a high Volatile Organic Compound (VOC) content.

- Standard System Impact: Traditional solvent-borne coatings can have VOC levels from 250 to 450 g/L. VOCs are precursors to smog and can impact air quality.
- Low-Emission Alternatives:
 1. Low/Zero-VOC Coatings: Specify paint systems with VOC content <100 g/L. Many high-performance, water-based, and high-solids formulations are available.
 2. Powder Coating: A zero-VOC option, though typically limited to smaller components due to oven size constraints.
 3. Hot-Dip Galvanizing (HDG): An alternative corrosion protection system with extremely high durability and zero on-site VOC emissions. Its embodied carbon should be evaluated separately if considered.

Recommendation: Require submittals for coating systems to include VOC content data. Specify a maximum allowable VOC limit (e.g., <100 g/L) in the project documents.

5. Material Efficiency & Waste Factor

During fabrication, waste is generated from cutting members to size and from remnants (off-cuts).

- Estimated Waste Factor: 8-12% is a typical range for structural steel fabrication.
- Project Waste Estimate: 220 tons * 10% = 22 tons of scrap steel.
- Circularity: A key advantage of steel is that this waste is not landfilled. It is 100% recyclable and has high economic value. The 22 tons of scrap will be collected and sold back to a steel mill, typically an EAF, directly re-entering the supply chain. This creates a highly effective circular economy loop.

Recommendation: Encourage early collaboration with the fabricator to optimize material usage through design detailing (e.g., minimizing cuts, using standard stock lengths) to reduce initial waste generation.

6. EPD / LEED / BREEAM Contribution Summary

This steel package can make a significant and direct contribution to green building certifications.

- EPD (Environmental Product Declaration): Request a mill-specific, product-specific Type III EPD. This is the official documentation verifying the embodied carbon, recycled content, and other environmental impacts. Most major EAF mills in North America provide EPDs.
- LEED v4.1 Contribution (MR Category: Materials and Resources):
 - MRC: Building Product Disclosure and Optimization – EPDs:
 - Option 1 (1 point): Achievable by sourcing from a manufacturer (the steel mill) that provides a product-specific EPD.
 - Option 2 (1 point): Potentially achievable. Requires the sourced EAF steel's EPD to demonstrate a Global Warming Potential (GWP) lower than the industry-average baseline. EAF steel is highly likely to meet this.
 - MRC: Building Product Disclosure and Optimization – Sourcing of Raw Materials:
 - Option 2 (1 point): Achievable. This credit rewards products meeting high recycled content thresholds. The >90% recycled content of EAF steel easily qualifies for this credit.

LEED Summary: By specifying and procuring EAF-produced structural steel with a qualifying EPD, this 220-ton package can directly contribute to achieving up to 3 points under the LEED v4.1 MR category.

Disclaimer:* The carbon factors and estimates provided are based on industry averages and established databases for preliminary analysis. Final values should be confirmed using project-specific Environmental Product Declarations (EPDs) from the selected steel mill and fabricator.